AUR
A

AUGMENTED USER ROAD ASSIST

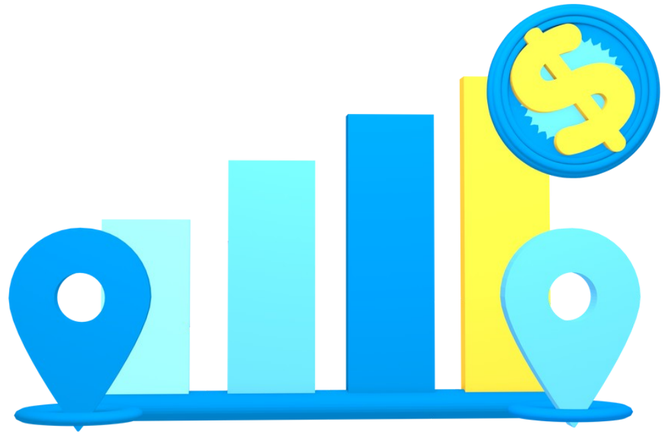


“AR - DRIVEN REAL TIME DRIVING ASSISTANCE & AI COMPANION”

Created by Team AutoNova

AUR
A

Table of Contents



Our Team | Background | Problem Statement | Approach

Process Flow | Technical Details | Code Snippets | Visual Graphics

Challenges Faced | Achievements | Future Enhancements | Project Plan

AUR
A

OUR TEAM



Razzaq Ahamed Shajahan



Rajesh S



Tanya V



Navitha M

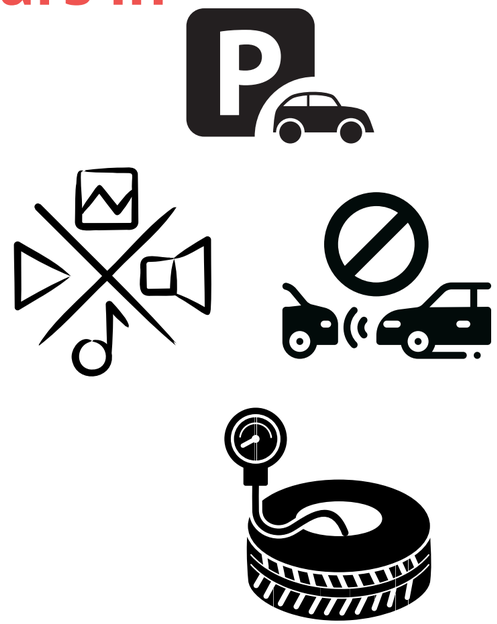


Monika E

AURA

BACKGROUND

What features do current infotainment systems in passenger cars in India offer ?



- Touchscreen Display – Central control for music, calls & settings
- Apple CarPlay / Android Auto – Seamless smartphone integration
- GPS Navigation – Real-time directions with traffic updates
- Bluetooth & Voice Control – Hands-free calls and media commands
- Tire Pressure Monitoring System (TPMS) – Alerts when tire pressure is low.
- Vehicle Alerts & Controls – AC, reverse camera, and system notifications

AUR

PROBLEM STATEMENT

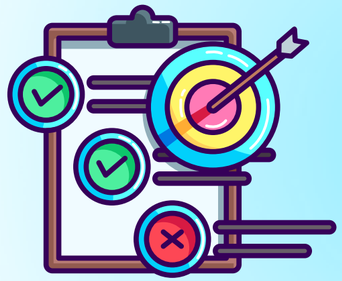


What they don`t offer

- No driver fatigue detection - no real time suggestions for taking breaks
- Lack of Google Calendar based day planner with voice assist - listing out the appointments for the day
- Absence of Weather Alerts - Suggestions for planning trip route in case if heavy rains enroute or near destination

OBJECTIVES

- To develop an Al + AR-based smart driver assist.
- To create AI based Google Calendar integrated day planner / upcoming trip planner
- To provide real-time alerts and navigation support



AUR

APPROACH

AR INTEGRATION

For Al + AR-based smart driver assist.

AI VOICE COMPANION

For AI based Google Calendar integrated day planner / upcoming trip planner

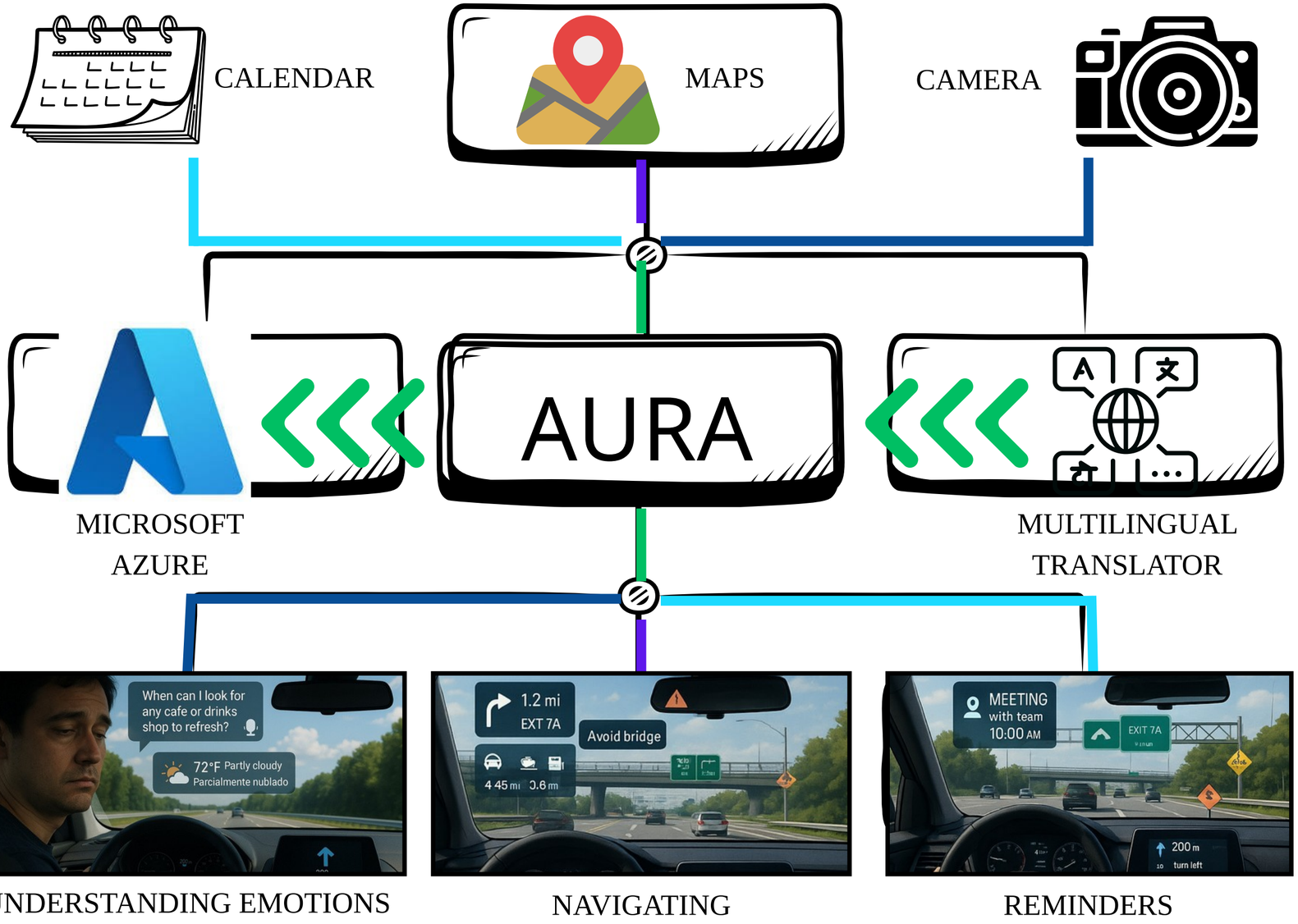
MICROSOFT AZURE

For real-time alerts and navigation support



AUR
A

PROCESS FLOW



TECHNICAL IMPLEMENTATION

- Unity 3D + AR Foundation
  - In-car AR visual interface
- Google Calendar API + Azure App Service
  - Event & reminder sync
- IR Camera + Azure Face API
  - Emotion & drowsiness detection
- Azure Maps APIs
  - Live routing & POI suggestions
- Azure Speech Services
  - Natural voice input/output
- Multilingual Support
  - Tamil, Hindi, English (auto-detect)
- Cosmos DB + Blob Storage
  - Store preferences, logs, media
- Azure Functions
  - Automate alerts & backend logic



AUR

🔧 CODE SNIPPETS BY MODULE

A 🗣️ SPEECH RECOGNITION + TTS (MULTILINGUAL)

```
import azure.cognitiveservices.speech as speechsdk

speech_config = speechsdk.SpeechConfig(subscription="YOUR_KEY",
region="YOUR_REGION")
speech_config.speech_recognition_language = "ta-IN"
speech_config.speech_synthesis_voice_name = "ta-IN-PallaviNeural"
recognizer = speechsdk.SpeechRecognizer(speech_config=speech_config)
synthesizer = speechsdk.SpeechSynthesizer(speech_config=speech_config)
print("🎤 Speak in Tamil:")
result = recognizer.recognize_once()
print("You said:", result.text)
synthesizer.speak_text_async("உங்களுக்கு அருகிலுள்ள உணவகம் உள்ளது").get()
```

🗺️ AZURE MAPS NEARBY POI SEARCH

```
import requests

url = "https://atlas.microsoft.com/search/poi/json‘
params = { "subscription-key": "YOUR_AZURE_MAPS_KEY", "api-version": "1.0",
   "query": "restaurant", "lat": 13.0827,   # Example: Chennai coordinates
   "lon": 80.2707, "radius": 5000}
response = requests.get(url, params=params)
data = response.json()
for poi in data['results']:
   print("Found:", poi['poi']['name'])
```

AURA

🔧 CODE SNIPPETS BY MODULE

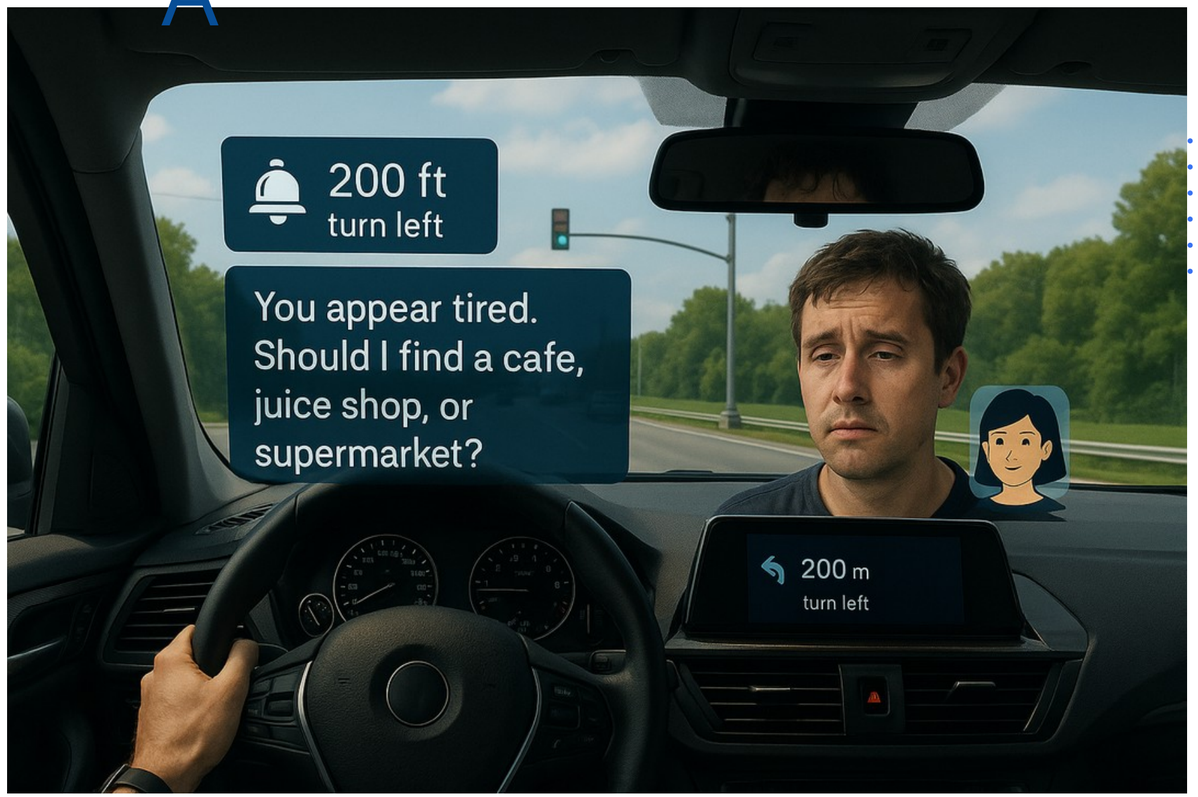
DRIVER EMOTION DETECTION (USING AZURE FACE API)

```
import requests

image_path = "driver.jpg"  # Frame from IR camera
with open(image_path, "rb") as image:
    response = requests.post(
        "https://<your_region>.api.cognitive.microsoft.com/face/v1.0/detect",
        params={"returnFaceAttributes": "emotion,headPose"},
        headers={"Ocp-Apim-Subscription-Key": "YOUR_FACE_API_KEY",
                 "Content-Type": "application/octet-stream"},
       data=image.read() )
faces = response.json()
for face in faces:
    emotions = face["faceAttributes"]["emotion"]
    print("Emotion Scores:", emotions)
```

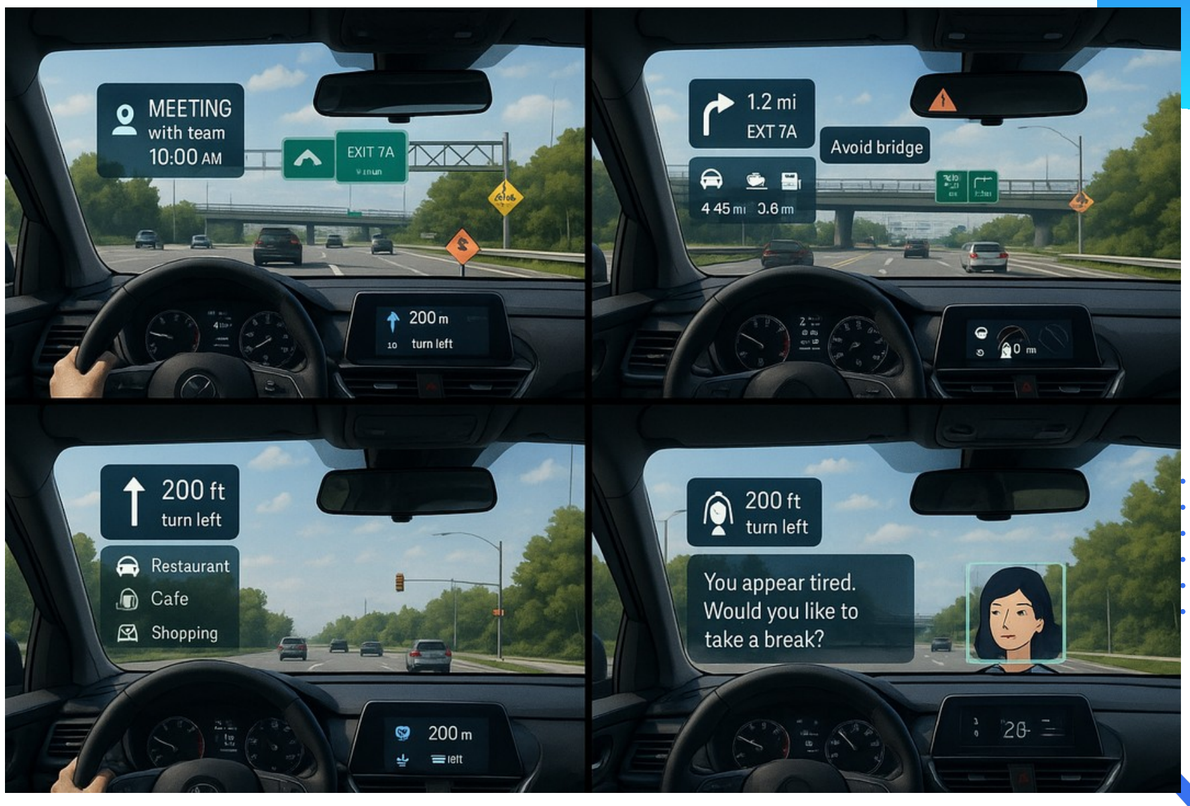
AUR
A

SIMULATED VISUAL GRAPHICS



READING FACIAL EXPRESSIONS AND GIVING SUITABLE RECOMMENDATIONS

VOICE ALERTS FOR REMINDERS & NAVIGATION



AUR

A

PLAN TO PROTOTYPE : WHAT WE FACED

EXPECTED CHALLENGES	
Hardware Integration(HSI)	
Processing Real-Time Data(PRTD)	Manu... spee... light...
Driver Distraction & Safety(DDS)	AR ov... driv...



ACHIEVEMENTS - RECOGNITION & USER FEEDBACK



First-Time submission

- This is the initial version of the project, being presented for the first time at TATA InnoVent.

Prototype - Under development

- Core features including AR navigation, voice interaction, and emotion detection are in the development and testing phase.

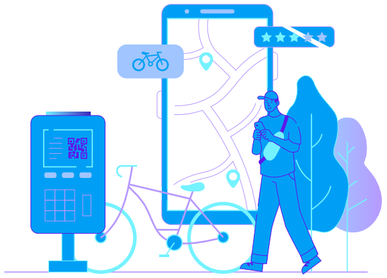
User testing and feedback- yet to be collected

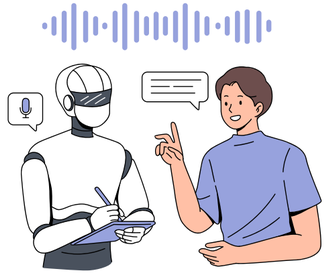
- User trials and feedback collection are planned for the next development phase after feature integration.

AUR

A Future Enhancement Ideas

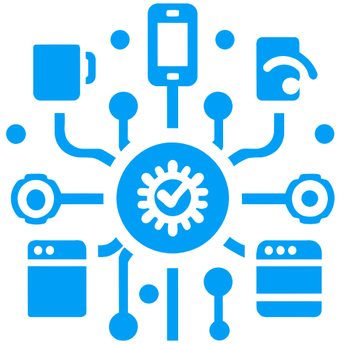
AUGMENTED USER ROAD ASSIST for real-time alerts, navigation, and voice interaction - specifically tailored for **motorcycles** and **scooters**.





- AR Helmet HUD (Heads-Up Display)
- Voice Assistant Integration
- Smartphone Integration
- Sensors & Connectivity





📈 PROJECT PLAN (3 MONTHS)

- **PHASE 1** (WEEKS 1–4)

Goal: Core prototype — greeting, playlist, calendar

Tools: Raspberry Pi, Android app

- **PHASE 2** (WEEKS 5–6)

Add: Route tracking + basic voice assistant

Tech: Google Maps API, TTS

- **PHASE 3** (WEEKS 7–8)

AI: Emotion detection (fatigue/yawn)

Tools: OpenCV, TensorFlow Lite

- **PHASE 4** (WEEKS 9–10)

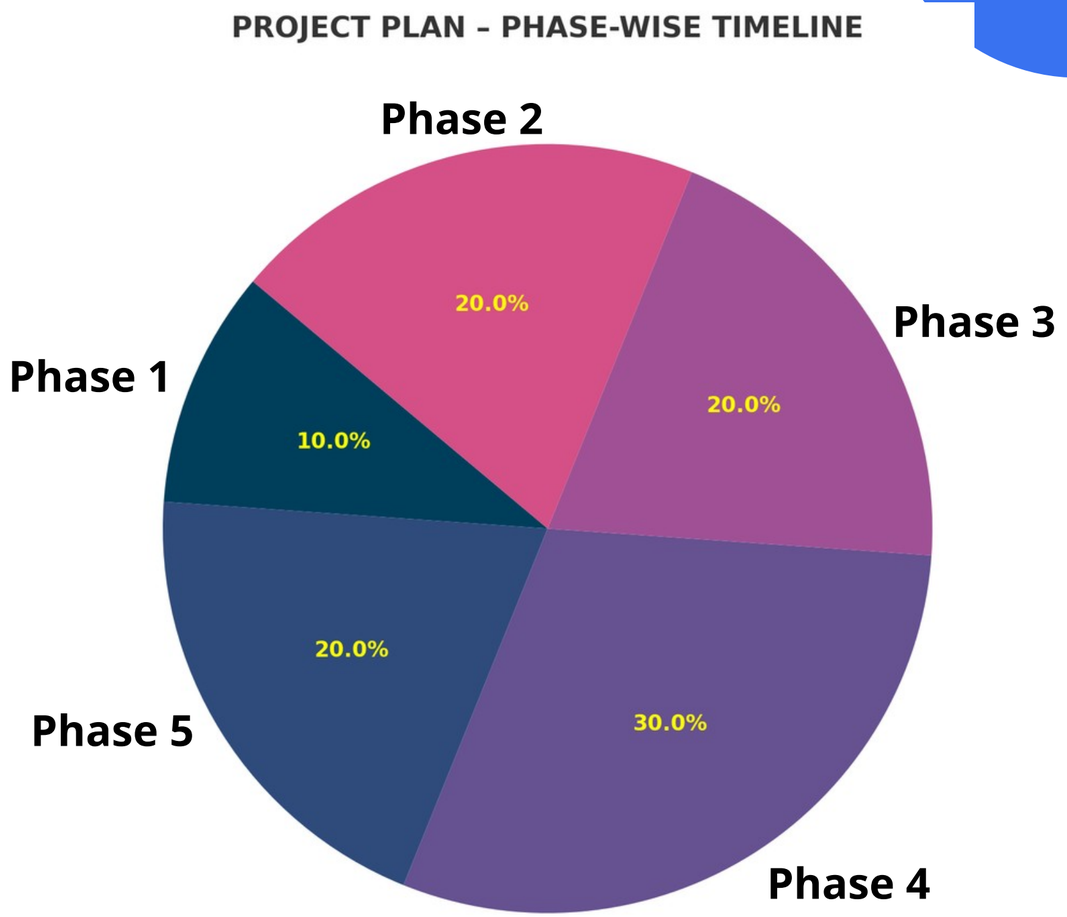
App: Companion app(development) + cloud sync

Backend: Firebase

- **PHASE 5** (WEEKS 11–12)

Test: Real-time drive demo + report

Output: Final project + viva-ready



AUR
A

THANK YOU!!

